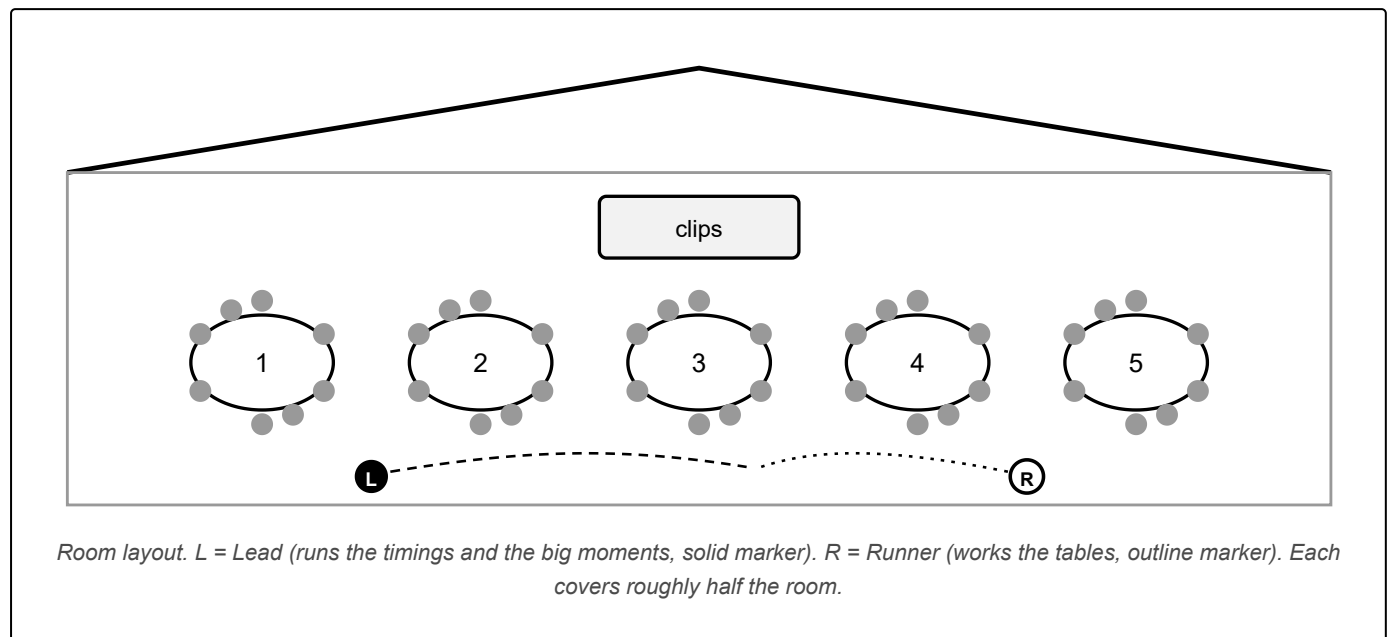


Who lives on a seaweed farm?

A one hour, hands-on workshop for adults. Build the food web of a seaweed and shellfish farm on the table in front of you, then take the farm away and watch what happens.

60	30-40	5	2
minutes	participants	tables of 8	facilitators



• THE ONE THING TO GET RIGHT

Everything builds towards minute 24, when you physically remove the farm. Protect that moment. If you are running late, cut the discussion, never the removal.



KIT LIST

- ☐ 5 × A1 laminated farm mats
- ☐ 5 × deck of 8 species cards (40 total)
- ☐ 1 spare deck (for a sixth table or losses)
- ☐ 12 × wipe-clean pens, plus cloths
- ☐ Flipchart pad and 4 chunky markers
- ☐ Laptop, charged, clips downloaded offline
- ☐ Printed stills of the 3 clips as backup
- ☐ Mussel shells, a metre of dropper rope
- ☐ Fresh kelp, collected that morning
- ☐ 2 × A5 crib sheets (page 9)

▲ WATCH OUT

Test the pens on the laminate before the day. The wrong marker is permanent and you will destroy £100 of mats in the first ten minutes. Buy wipe-clean, not dry-wipe-adjacent, and take a cloth per table.

THE STORY YOU ARE TELLING

BEAT	WHAT THE ROOM LEARNS
Build the web	A seaweed and shellfish farm is not empty water. It is crowded.
Remove the farm	A third of that life was only there because of the farm.
The sea potato	It is not all good news, and we say so first.
Costs and gives	This is a trade-off worth measuring, which is why PEBL films it.

• THE NUMBERS, IF YOU ARE ASKED

Across the full catalogue we map 72 species and 238 feeding links on a farm. Take the farm away and it is 51 species and 151 links. The farm builds a bit under a third of the species and just over a third of the feeding connections. Say "roughly a third" out loud, not the precise figures.

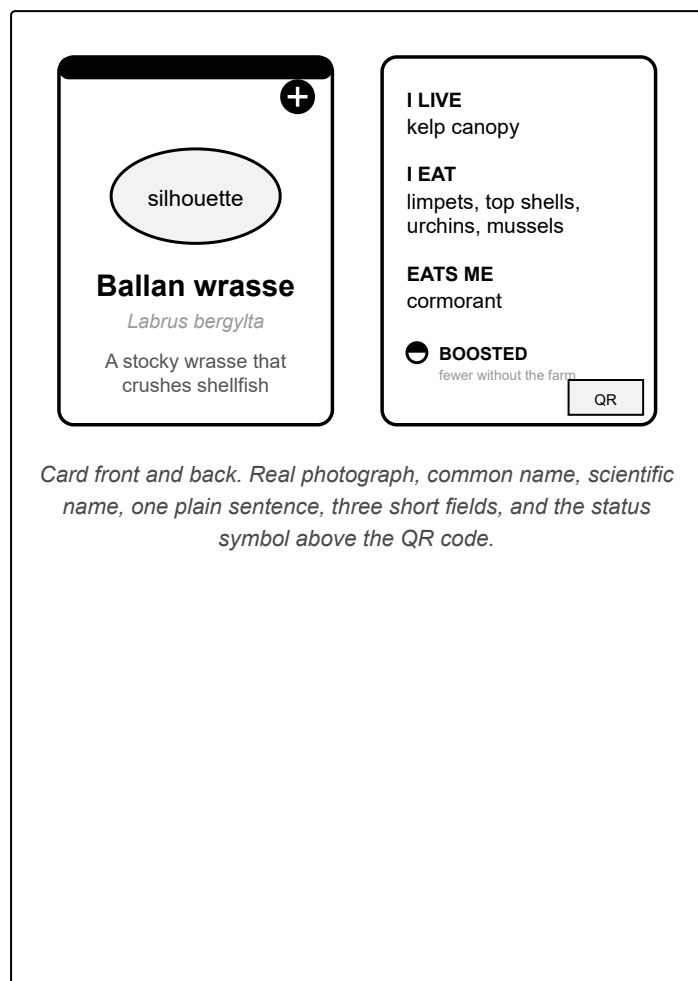
▲ WE COUNT CONSERVATIVELY, AND YOU SHOULD SAY SO

"Farm-built" is only used for animals that physically *cannot* live on bare sand — things that need a hard surface to sit on or a crevice to hide in. Big mobile animals that the farm merely draws in from the surrounding seabed (wrasse, cod, pollack, octopus) are counted as boosted, not built. If someone asks whether you are marking your own homework, that distinction is your answer. Page 10 explains it properly.

WHAT TO PRINT

ITEM	SPEC	COST
Farm mats ×5	A1, colour, encapsulated (gloss). Farm cross-section with the four food sources printed on it, no animals	~£100
Species cards ×80	A6, 350gsm, double sided, colour. Two identical sets of 40 so the activity survives being given away	~£90
Wipe pens, cloths	Fine tip, non-permanent	~£15
Flipchart and markers	Plus tape for the wall	~£25
Props and contingency	Shells, rope offcut, kelp on the day	~£25
Total		~£255

The mats and species cards above are the only colour items — they go to a proper print shop. This guide and the evidence cards are built for an ordinary black & white office printer or photocopier.



CARD RULES

- **Name at 14pt minimum.** A marquee is dim and half your room is over 60.
- **Scientific name in grey italics** under the English, for anyone who knows their species.
- **One plain sentence**, so nobody has to admit they have never heard of a dragonet.
- **Never colour alone.** Each status gets a symbol and a word, not a colour.
- **Max three fields on the back.** If it needs a fourth, cut it.
- **QR on every card**, to FishSpotter. The card is the takeaway, so there is no leaflet.

THE FOUR SYMBOLS

- **FARM-BUILT** ×15 Could not live here at all
- **BOOSTED** ×8 Here anyway, drawn in or fed
- **HERE ANYWAY** ×16 A sand or open-water native
- **BETTER WITHOUT** ×1 Does worse under a farm

• BUILT VERSUS BOOSTED

Only animals that cannot live on bare sand at all get the filled circle. Big mobile visitors that come to the farm to feed — wrasse, cod, pollack, octopus — are boosted at most. Page 10 explains it properly.

Do not laminate the species cards. People take them home and a laminated card feels like something to hand back.
350gsm card is warmer, cheaper, and posts through a letterbox.

People are still arriving. Do not wait for them. Start with the clips playing and talk over the top.

Play three short clips filmed on a working Welsh farm: a pollack, a spider crab on a rope, a cuttlefish. Ask the room to call out what they can see. Take every answer, including the wrong ones.

“ SAY ROUGHLY THIS

"Everything you have just watched was filmed on a seaweed and shellfish farm, about an hour up the coast. Nobody put those animals there. They turned up."

▲ GET IN FRONT OF THE SALMON QUESTION

Somebody in this room thinks a marine farm means caged salmon, sea lice and chemicals. If you let that sit unspoken you will spend the hour under suspicion. Say it yourself, now, before anyone asks:

"Quick thing, because people always ask. Nothing here is fed, caged or medicated. Seaweed and mussels take food out of the water rather than having it poured in. It is the opposite of a salmon farm. If you want to get into that, grab me at the end."

Runner deals one deck per table, face up, and one mat per table. Everyone takes a card. Late arrivals slot into any table with a spare card.

“ SAY ROUGHLY THIS

"Take a card. You are now in charge of that species. Have a read. You need to know three things: where you live, what you eat, and what eats you."

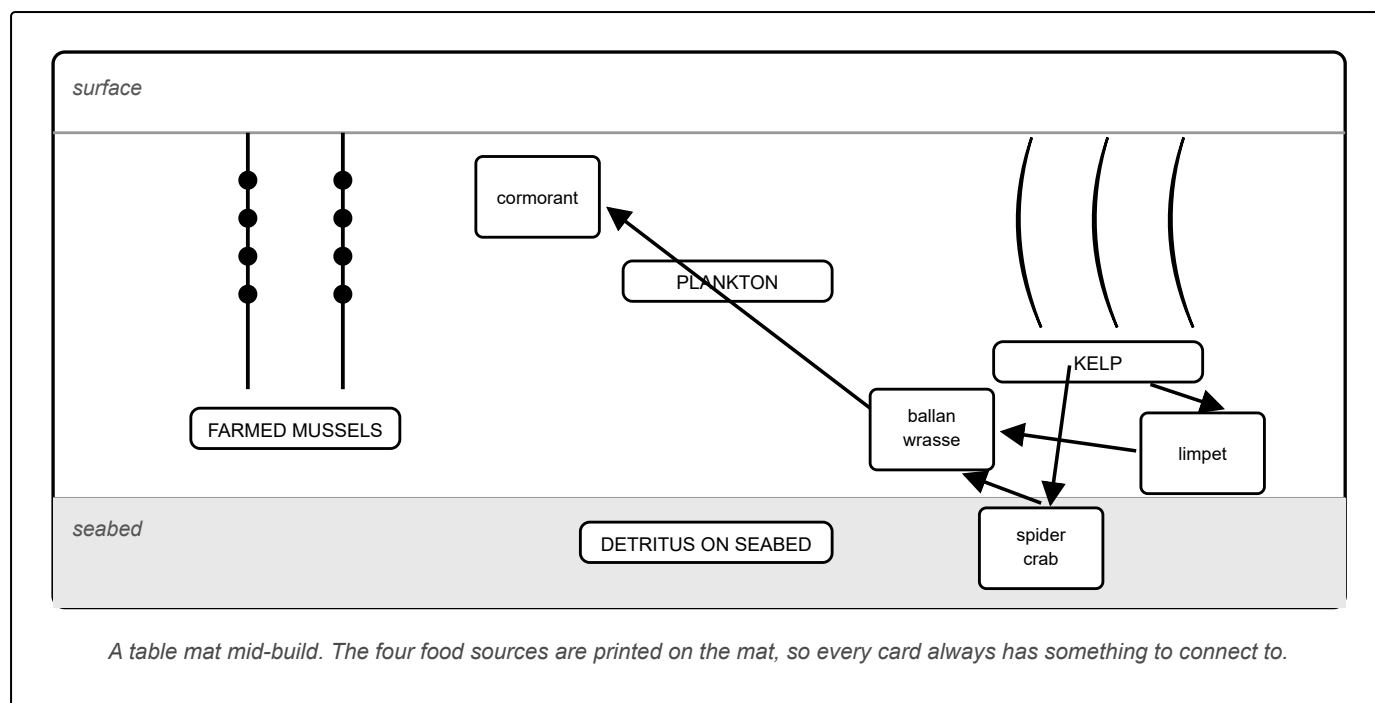
• WORDING MATTERS

Say "you are in charge of the cuttlefish", not "you are a cuttlefish". Adults will happily be an expert on an animal. They will not happily pretend to be one, and the difference costs you nothing.

• IF SOMEONE WILL NOT TAKE A CARD

Fine. Give them the job of counting arrows for the table. They are now the most useful person there, and they will usually pick up a card within ten minutes anyway.

Tables place their cards on the mat by where the species lives, then draw an arrow from each meal to whoever eats it. Arrows point the way the food travels.



• WHY THE DECKS ARE PRE-SORTED

Each table has been given a set of eight species that genuinely eat each other. Do not shuffle the decks together or let tables swap cards. If you do, tables end up drawing arrows to species that are sitting on a different table, and the activity stalls.

PROMPTS THAT UNSTICK A TABLE

- "Start with the food sources printed on the mat. Who eats straight off those?"
- "Right, who at this table would eat *you*?"
- "Follow it up. Who eats the thing that eats you?"

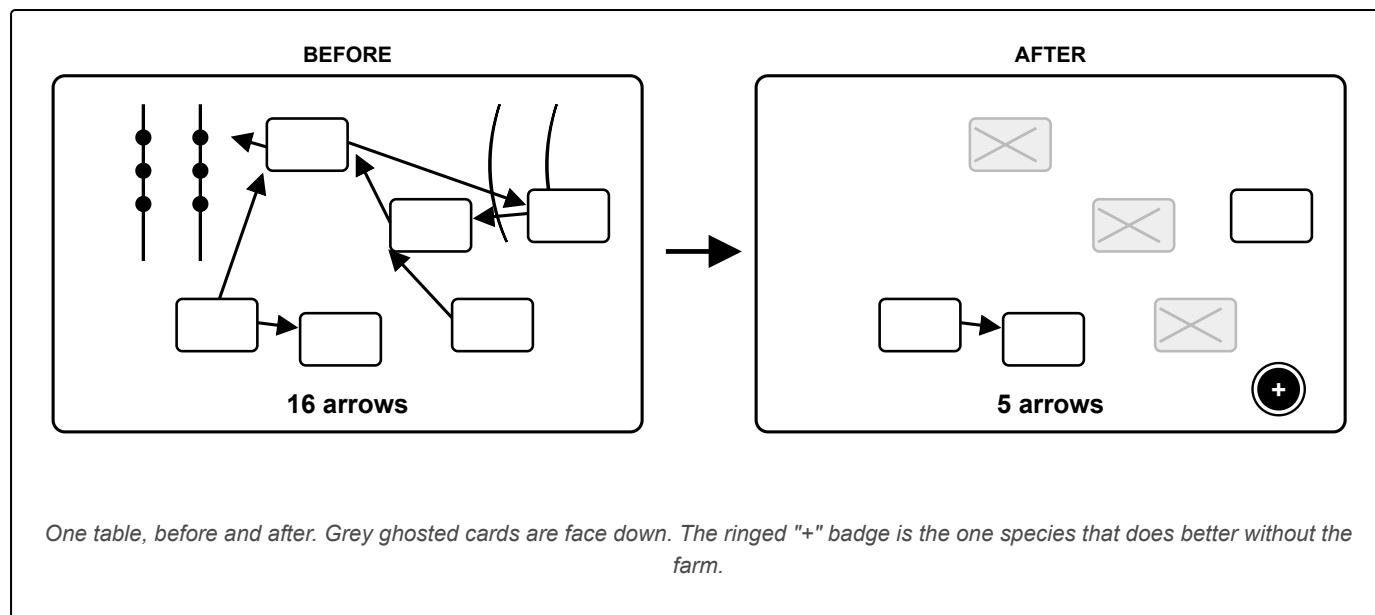
PACE

- **Fast table:** "Count your arrows and write the number in the corner. Now find the longest chain you can."
- **Stalled table:** sit down with them and place two cards yourself. Do not narrate, just start.

▲ EVERYONE MUST WRITE THEIR ARROW COUNT ON THE MAT BEFORE MINUTE 24.

You need that number for the reveal. Runner walks the room at minute 22 and checks every table has one.

This is the whole workshop. Ninety seconds of it. Direct it properly: one voice, everyone quiet, no side conversations.



RUN IT IN THIS ORDER

1. **Stop the room.** "Pens down, everyone look up." Wait for actual quiet.
2. **Both facilitators move in.** Lead counts down from three. On "one", both of you physically lift the kelp and the mussel rope props off the tables and put them on the floor, out of sight.
3. **Beat of silence.** Do not fill it.
4. "If your card has the filled circle, the farm was the only reason you were here. **Turn your card face down.**"
5. "Now cross out every arrow that touches a face-down card."
6. **Go round the tables out loud.** "Table one, what were you and what are you now?" Let them say the numbers themselves. Table 1 should report 16 down to 5; page 10 has the rest.
7. **Then the star card.** "Is anyone better off?" Table 5 has the sea potato. Get that person to hold it up.

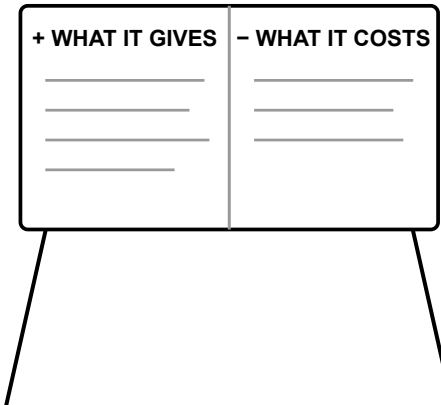
• WHY TABLE 5 MATTERS

The tables are deliberately uneven. Table 1 loses four of eight, and since those four are the bottom of its food chain it is close to wiped out. Table 5 loses one and gains one that does better without the farm. That spread is the honest picture, and the room notices it without you saying it.

▲ DO NOT RUSH TO THE MORAL.

The temptation is to explain what just happened. Do not. Ask "so what did that just show you?" and let a member of the public say it out loud instead. It lands ten times harder in their words.

Two columns on the flipchart. The room fills them in. Your job is to write, not to argue.



Keep it on one sheet so the whole room can see both columns at once.

WHAT USUALLY COMES UP

Gives. Shelter and nursery space for young fish. Food for crab and lobster. Somewhere for things to settle. Calmer water inside the lines. Food produced without feed, fresh water or fertiliser.

Costs. Sea space that other people used. Gear and navigation conflict. Enriched mud under the lines that suits fewer clean-sand species. The habitat disappears when the crop is harvested. Ropes and floats need maintaining or they become litter.

• HAND THE PEN TO THE SCEPTIC

If there is a fisherman in the room, and at this festival there will be, invite them to fill in the costs column themselves. It is the single highest value thing you can do all hour. It turns the person most likely to dismiss you into the person who helped write the honest version, and everyone else sees you do it.

“ SAY ROUGHLY THIS TO CLOSE THE STEP

"We are not here to tell you a farm is all upside. We are here because nobody has actually measured this properly in Wales, and that is the job we are doing. Both columns are real."

▲ DO NOT LET THIS BECOME A PLANNING MEETING.

If two people start relitigating a specific local licence application, take it offline: "That is a proper conversation and I want to have it, can we do it at the end so we get through the rest?" Then genuinely do it at the end.

Once both columns are full, hand each table its **evidence card** from the separate pack. One person per table reads the headline out loud. Three minutes, and it changes the register of the room completely, because the session stops being your opinion and starts being a measurement.

TABLE	THEIR CARD	THE FINDING
2 ropes	The reef that came back after 150 years	30% more reef in 7 years, Lyme Bay
3 shoal	More fish, and young ones settling in	1.6× more fish, larvae of 7 species
4 seabed	Crab and lobster do well out of it	Commercial crustaceans gain
1 canopy	A patch of easier water for shellfish	23% easier shell-building
5 sand	A farm is not a wild kelp forest	Fewer species than natural habitat

MAKE THEM GUESS FIRST

Before table 2 reads their number, stop them and ask the whole room:

“ SAY ROUGHLY THIS

"They watched the seabed under a mussel farm in Lyme Bay for eight years, against control sites nearby. Hands up who thinks the seabed got better. Right, how much more reef after seven years? Shout me a percentage."

Take three guesses, then let table 2 read out **30%**. A number you have guessed at is a number you remember. This is the cheapest engagement trick in the guide and it works on adults every time.

• READ TABLE 5'S CARD LAST, AND DO NOT SOFTEN IT

There is the study that found kelp farms hold *fewer* species than natural habitat nearby. Ending the evidence round on the finding that works against you is the strongest thing you will do all hour. Follow it with: "which is why we think farms belong on bare ground, not on top of something that is already good."

▲ TWO CARDS ARE YOURS, NOT THE TABLES'

The carbon card and the PEBL card stay with the facilitator. Use the carbon card the moment anyone mentions carbon, and the PEBL card to close the session.

Print this page at A5 and carry one each. These four come up nearly every time.

1. "ISN'T THIS THE SAME AS A SALMON FARM?"

No, and it is the opposite in one specific way. Salmon are fed, so a salmon farm puts food, nutrients and sometimes medicines into the water. Seaweed and mussels are not fed anything. They take nutrients out of the water to grow. Different industry, different footprint, and it is fair to judge them separately.

2. "DOESN'T SEAWEED LOCK UP LOADS OF CARBON?"

It grows fast and it absorbs carbon dioxide while it grows. But if we harvest it and eat it, or put it on a field, or in a bottle of shampoo, that carbon comes back out fairly quickly. Only the fraction that sinks to deep water and stays there counts as storage, and nobody has measured that well yet for UK farms. The strong climate case here is habitat and shelter, not storage.

• SAY THIS EVEN WHEN NOBODY ASKS

Somebody in the room will confidently tell their neighbour that seaweed farms are a big carbon solution. Correcting it gently, in public and early, is the single biggest credibility win available to you. People trust an organisation that argues against its own hype.

3. "THAT'S MY FISHING GROUND YOU ARE COVERING UP."

Straight answer: yes, a farm takes sea space, and that is a genuine cost to somebody. What we can say is that the structure feeds and shelters things that crab and lobster eat, and young fish use it as cover. Whether that is worth the space you lose depends on where it goes and how big it is, which is exactly why we put cameras on them instead of guessing.

▲ DO NOT OVERSELL TO A FISHERMAN.

They will know more than you about that stretch of coast. Ask where they work and what they see on their gear. Treat it as data collection, because it is.

4. "WILL THIS HELP WITH THE SEA WARMING UP?"

Not the temperature itself. It helps things cope. The lines take the energy out of waves so there is calmer water inside. Seaweed photosynthesising raises the pH of the water right around it, which matters because the sea is slowly acidifying and that makes it harder for young mussels, crabs and other shellfish to build their shells. A farm is a small patch of easier conditions.

5. "WHO PAYS YOU?"

PEBL is a community interest company, not a private business with shareholders. Say who actually funded the work, name them, and move on. Evasiveness here costs you the room.

Each table gets eight cards that genuinely eat one another. Keep the decks separate. ● farm-built, ① boosted, ○ here anyway, ♦ better without. The next ten pages are the full answer sheet: for each table, the finished mat with every arrow drawn, then the biology behind it.

TABLE	FACE DOWN	ARROWS	WHAT IT IS FOR
1 The kelp canopy	4 of 8	16 → 5	The wipeout. Its whole food chain stands on grazers that need a hard surface.
2 The mussel ropes	3 of 8	16 → 7	Competition. Six of eight cards eat the same meal.
3 The rope shoal	3 of 8	18 → 8	A clean four-step chain, and the clearest nursery story.
4 The seabed crew	4 of 8	17 → 5	Cover, not food. The bottom always had detritus; it lacked hiding places.
5 The open sand	1 of 8, and 1 gains	11 → 9	The control. Proof you are not overselling.

TWO THINGS A FARM DOES, AND ONLY ONE OF THEM COUNTS

This is the bit of marine biology worth carrying in your head, because every hard question in the session comes back to it.

WHAT THE FARM DOES	WHICH ANIMALS	HOW WE COUNT IT
Builds habitat	Anything that needs a hard surface to grip, or a crevice to hide in. Limpets, top shells, urchins, whelks, the big starfish, blennies, rock goby, conger eel.	● Farm-built. On bare sand there is genuinely nowhere for these to be. Take the structure away and they are not there.
Attracts visitors	Big, mobile animals with home ranges of hundreds of metres or more. Ballan wrasse, cod, pollack, octopus, eider, seals.	① Boosted at most. These already live in the surrounding sea. The farm is a feeding stop, not the reason they exist.

Building a house versus putting out a bird feeder. We only claim the houses. Four species that a looser reading would have called farm-built — ballan and cuckoo wrasse, and both octopuses — are deliberately counted as boosted for exactly this reason.

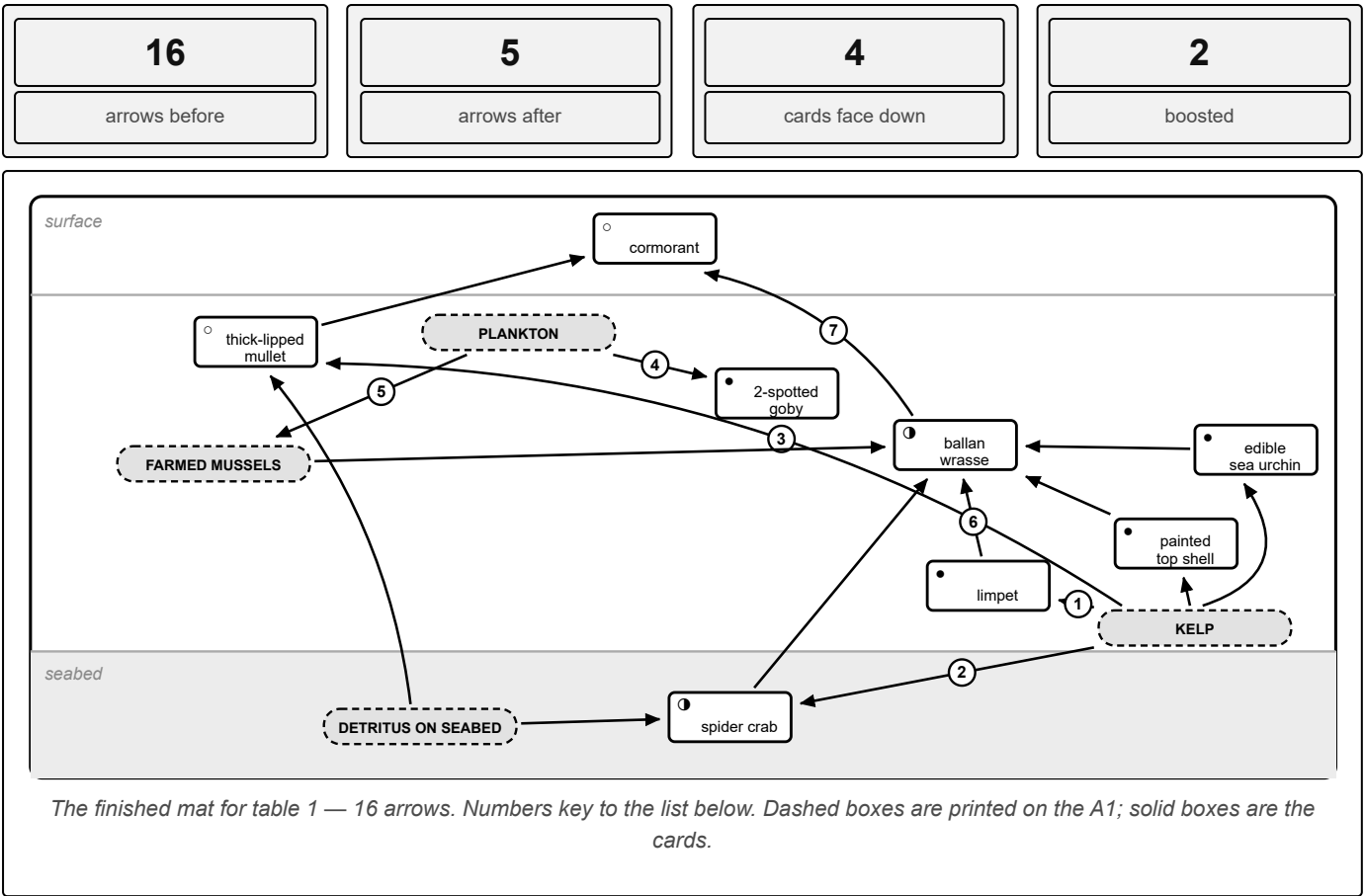
● DEAL TABLE 5 DELIBERATELY

It is the quietest table during the reveal and the most important one during the discussion. Put your most talkative or most sceptical group there, because they are the ones who will say "well, hang on, we barely changed" out loud, which is exactly the honesty the session needs.

▲ DO NOT READ THESE PAGES AT THE TABLES

They are for you, the night before. A facilitator reciting the correct answer while people are still placing cards kills the activity. Use them to check a finished mat, to answer "why is mine there?", and to keep your nerve when somebody who fishes for a living disagrees with you.

Seaweed feeds the grazers, the grazers feed the wrasse, the wrasse feeds the bird. Four clean steps, and the bottom two are built entirely out of animals that need something solid to hold onto.



THE ARROWS, AND WHY THEY EXIST

- ① The three grazers all eat the seaweed itself, or the algal film growing on it.
- ② Browses the weed and scavenges what has fallen. Two arrows in, which is what saves it.
- ③ Sieves the organic film off both the weed and the sediment surface.
- ④ A planktivore, despite living against the weed.
- ⑤ On every mat. The mussels filter plankton, which is why a farm adds no food to the sea.
- ⑥ Five arrows into one fish. Its throat teeth crush shell, so almost everything here is on its menu.
- ⑦ The bird takes the two biggest fish on the table.

WHERE EACH CARD GOES, AND WHY

CARD	PLACE IT	THE REASON
● Common limpet	On the kelp and ropes	A grazing snail that has to clamp onto something solid and scrape algae off the surface it sits on. Sand offers nothing to clamp to.
● Painted top shell	On the kelp	Same job as the limpet, and it climbs the fronds and stipes themselves. A hard-surface grazer with no hard surface has nowhere to be.
● Edible sea urchin	On the kelp and ropes	Britain's largest urchin and a heavy kelp grazer. Needs hard ground to hold station on while it feeds.
● Two-spotted goby	Hovering in the kelp	The odd one out among gobies: it hangs in midwater beside the weed instead of sitting on the bottom, and eats plankton. The weed is cover, not food — but no weed, no goby.
● Ballan wrasse	In the kelp	Eats the grazers, crushing shells with throat teeth, and raids the mussel ropes. Boosted, not built: adults hold territory over hundreds of metres of rocky ground and live on any nearby reef. The farm concentrates them.
● Spider crab	Kelp and seabed	Browses weed and scavenges the bottom. That second food source is precisely why it survives the reveal when the grazers around it do not.
○ Thick-lipped mullet	Near the surface	Grazes the film of algae and organic matter off surfaces and sediment. Moves freely between estuaries and open coast.
○ Great cormorant	At the surface	Dives for fish anywhere along this coast. The farm is one stop on a much larger circuit.

THE SAME MAT AFTER THE FARM GOES

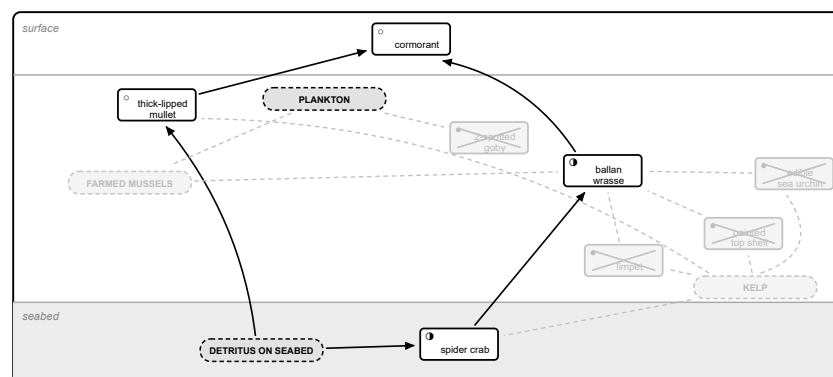
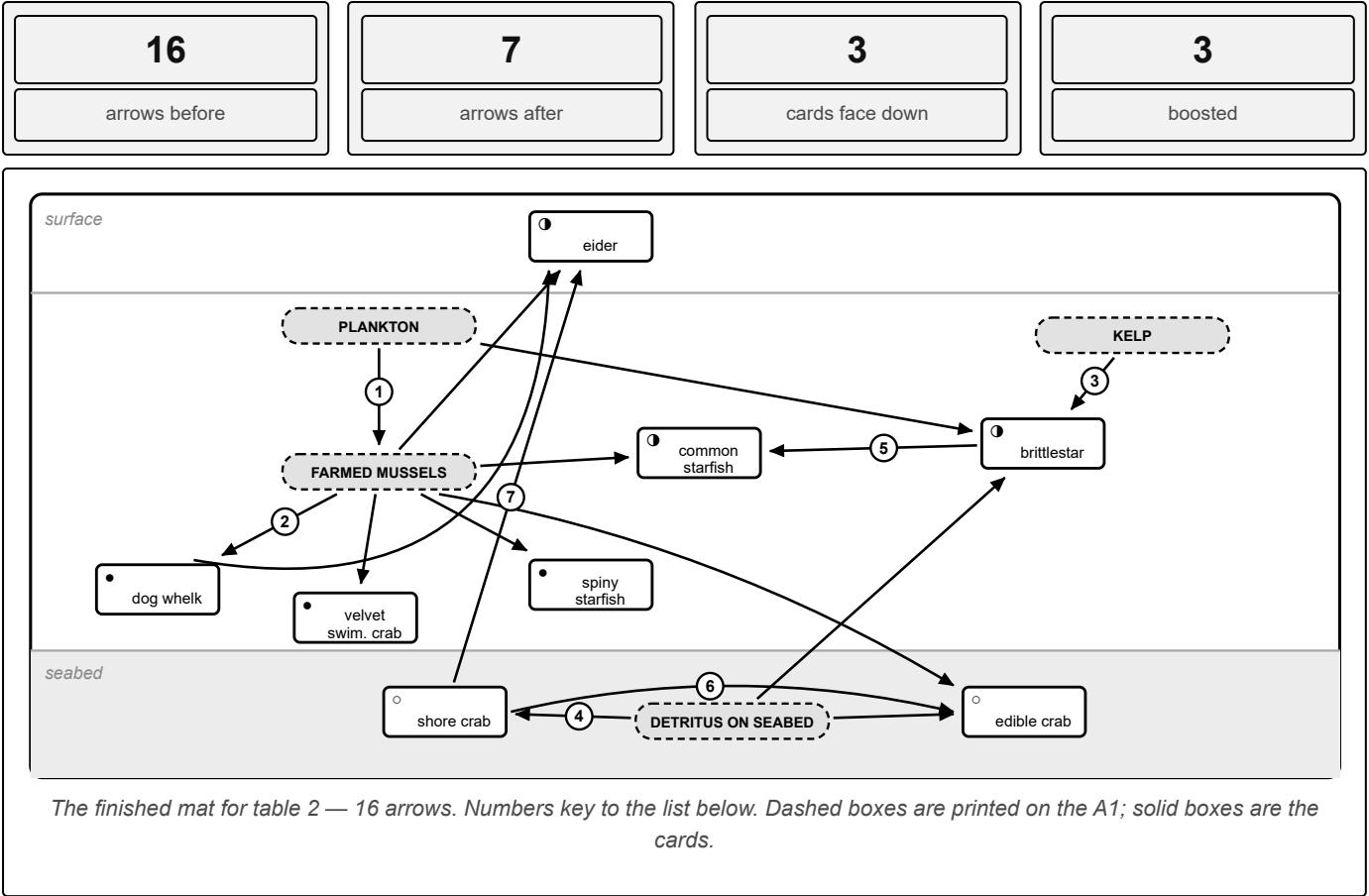


Table 1 after the reveal — 16 arrows down to 5. Crossed boxes are face down, faint dashed lines are the 11 arrows the table crosses out.

• THE TEACHING POINT

Lose the kelp and the ropes and you lose the entire bottom of the chain in one go, because every grazer here needs a hard surface. The wrasse and the cormorant technically survive, but the wrasse has lost four of its five meals. That is why this table drops from sixteen arrows to five while only turning over four cards — and it is the clearest illustration in the room that habitat matters more than headcount.

Six of the eight cards eat the same meal. This is the competition table, and the one where the honest answer is most mixed: three genuinely built, three boosted, two that could not care less.



THE ARROWS, AND WHY THEY EXIST

- ① On every mat, and here it is the start of everything: the mussels filter plankton.
- ② Six separate arrows off one food source. This is why a farm concentrates life: a lot of food in a small space.
- ③ It filters and deposit-feeds, taking whatever drifts past or settles. Three sources, so it is hard to starve.
- ④ Both crabs scavenge the seabed. This is their non-farm food, and it is why they survive.
- ⑤ The starfish's other meal. This single arrow is the reason it stays on the table.
- ⑥ Bigger crab eats smaller crab. People rarely expect this one.
- ⑦ Eider take far more than mussels — crabs and whelks go down whole too.

WHERE EACH CARD GOES, AND WHY

CARD	PLACE IT	THE REASON
• Dog whelk	On the ropes	Drills a neat hole through mussel and barnacle shell and rasps the animal out. Needs hard substrate to move over and shellfish to drill.
• Velvet swimming crab	On the ropes	A hard-ground crab with paddle-shaped back legs. It lives among rock and structure, and has little use for open sand.
• Spiny starfish	On the ropes	Britain's largest starfish and a bivalve specialist, found on rocky and mixed ground. The ropes are a vertical mussel bed.
❶ Common starfish	Ropes and seabed	Also a mussel specialist — but it occurs naturally right across sediment, and it eats brittlestars too, so it does not vanish. More of them with a farm, not none without.
❶ Common brittlestar	On the seabed	Forms dense natural beds on gravel anyway. Boosted because farm fallout feeds more of them. Three food arrows in, which is why it is the most robust card here.
○ Shore crab	On the seabed	Britain's commonest crab and a total generalist. Present on every shore, farm or no farm.
○ Edible crab	On the seabed	Generalist scavenger and predator that also takes smaller crabs. Mobile enough to be here regardless.
❶ Common eider	At the surface	Dives and swallows mussels, whelks and crabs whole. Boosted: eider are on this coast anyway, but a mussel farm is a reliable buffet and they raft up on them.

THE SAME MAT AFTER THE FARM GOES

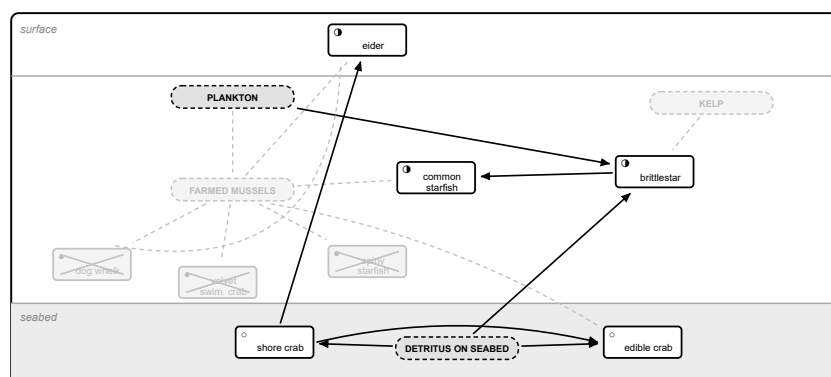


Table 2 after the reveal — 16 arrows down to 7. Crossed boxes are face down, faint dashed lines are the 9 arrows the table crosses out.

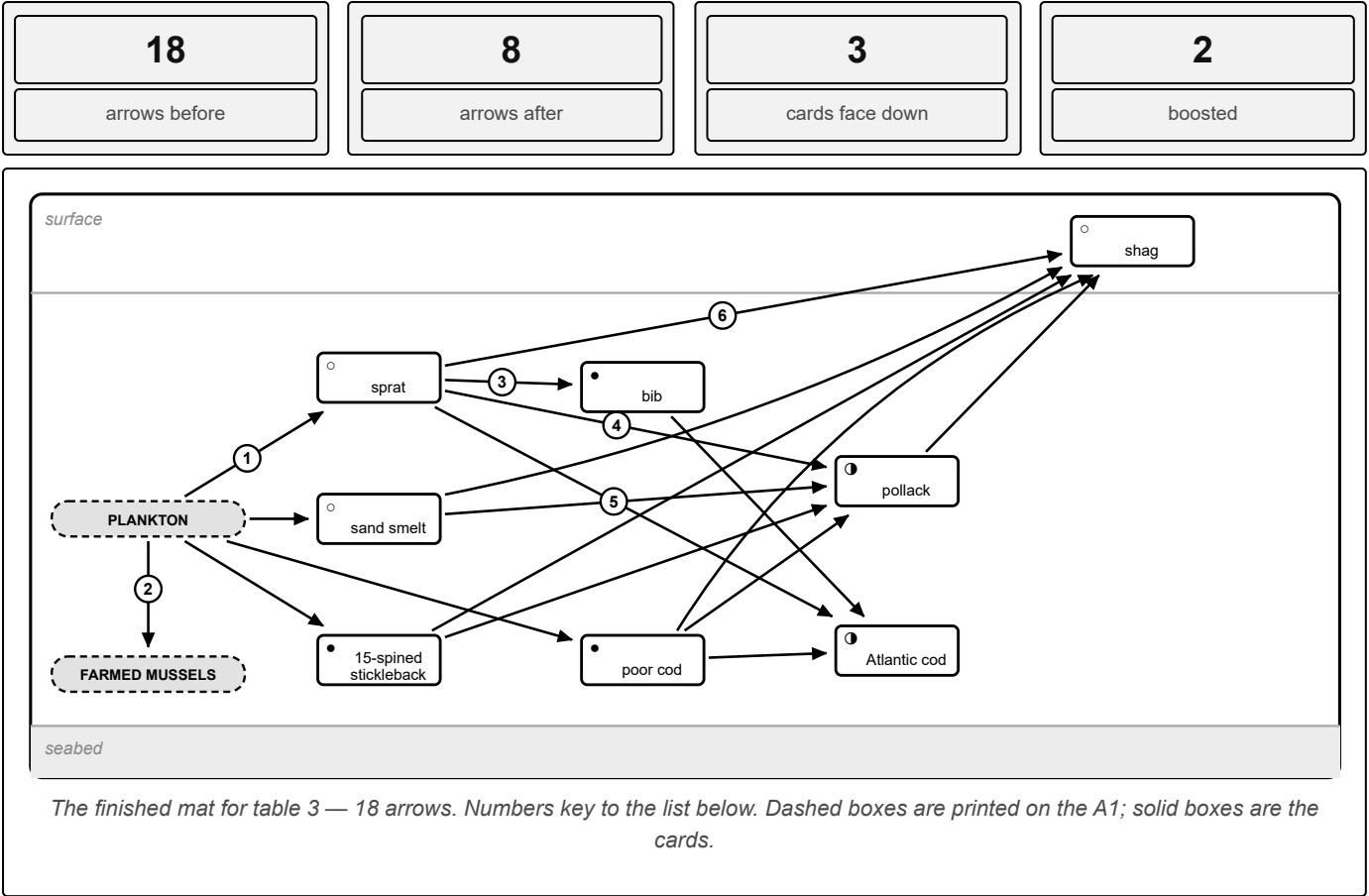
• THE TEACHING POINT

If a table argues the edible crab should be boosted rather than here-anyway,

• THEY ARE RIGHT AND YOU SHOULD SAY SO

— the research card table 4 reads out says commercial crustaceans do well out of farms. Conceding a point to a table is worth more than defending the pack. The wider lesson: this table barely halves its arrows, because most of its members had a second way to make a living.

Plankton to small fish to big fish to a diving bird. The cleanest food chain in the room, and the table where the farm's contribution is a nursery rather than a habitat.



WHERE EACH CARD GOES, AND WHY

CARD	PLACE IT	THE REASON
○ Sprat	Open water	Filters plankton in big shoals and drifts through. A farm does not make plankton, so the sprat neither needs it nor avoids it.
○ Sand smelt	Near the surface	Same story — a slender open-water planktivore that shoals near the top.
● Fifteen-spined stickleback	In the kelp	Hovers almost motionless among fronds, so well camouflaged it is nearly invisible. Totally weed-dependent, and a poor swimmer in open water.
● Poor cod	Around the ropes	A small gadoid that shoals tight to structure. See the caution below — this is one of the two least certain calls in the pack.
● Bib	Around the ropes	Same again: bib famously pack around wrecks, piers and any structure. Its young settle onto exactly this kind of habitat.
● Pollack	Midwater by the ropes	A mid-water hunter that comes in to work the shoal. Textbook attraction — it lives in the wider sea and visits to feed.
● Atlantic cod	Around the ropes	Same: a big mobile predator drawn in by the concentration of prey, not created by it.
○ European shag	At the surface	Dives for 20 to 40 seconds after small fish, anywhere on a rocky coast.

▲ POOR COD AND BIB ARE THE WEAKEST FARM-BUILT CALLS IN THE PACK — KNOW THIS BEFORE SOMEONE TELLS YOU

Both are mobile shoaling fish. An adult could perfectly well pass over open sand, so "gone without the farm" is a stronger claim here than it is for a limpet. They are marked farm-built because their *young* specifically settle onto structure, which is exactly what this table's own evidence card measures. If a biologist or a fisherman challenges it, concede straight away: say you would accept boosted, that the honest answer is somewhere between, and that this is precisely the thing PEBL is putting cameras on farms to settle.

THE SAME MAT AFTER THE FARM GOES

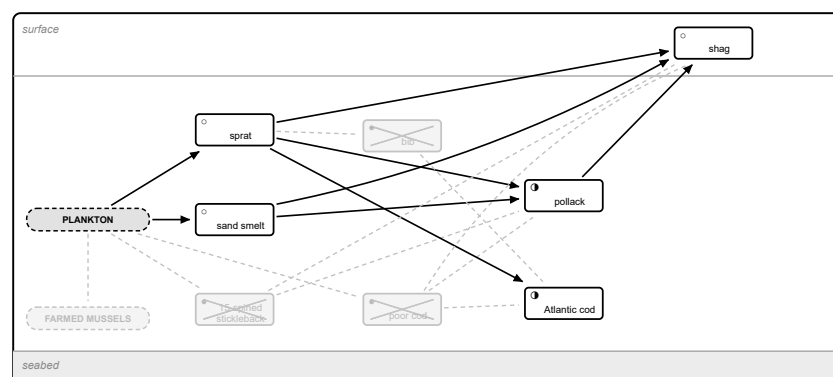
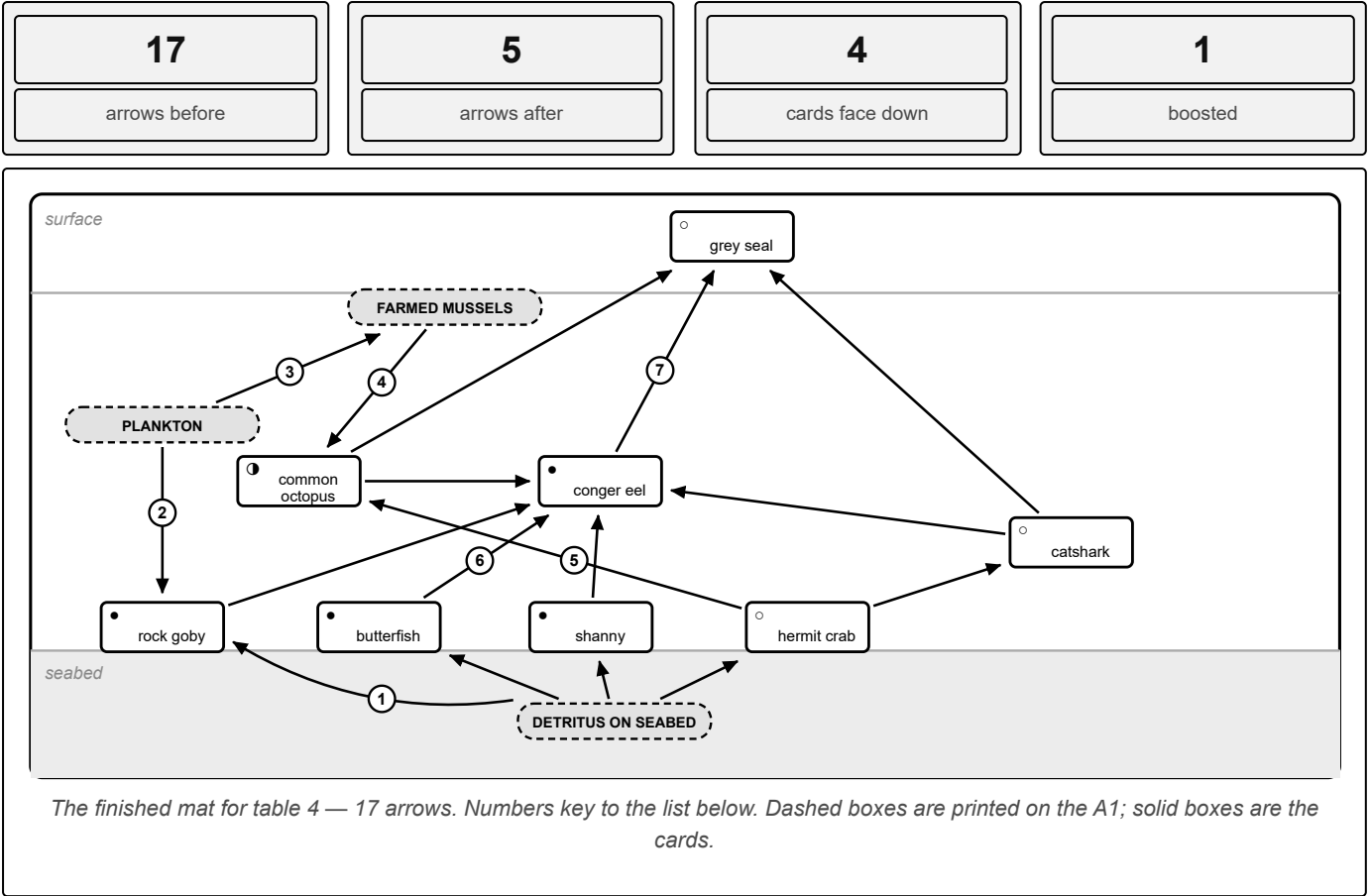


Table 3 after the reveal — 18 arrows down to 8. Crossed boxes are face down, faint dashed lines are the 10 arrows the table crosses out.

● THE TEACHING POINT

Nothing the farm does creates the plankton, so the base of this chain is untouched. What the farm adds is shelter for small fish to grow up in, and a reason for big fish to visit. Say that plainly: *"the farm didn't make the food here, it made somewhere to hide while you are small enough to be eaten."*

What falls off the farm rains down and feeds the bottom, and the anchor blocks give something big a place to live. This table is about cover, not food.



WHERE EACH CARD GOES, AND WHY

CARD	PLACE IT	THE REASON
○ Hermit crab	On the seabed	Scavenges detritus while wearing a borrowed whelk shell. A sand and mixed-ground native, entirely at home without a farm.
● Butterfish	Seabed, under cover	A ribbon-shaped fish that lives wedged under stones and among dropped shell.
● Shanny	Seabed, in crevices	A blenny that darts in and out of holes. Bare sediment gives it nothing to dart into.
● Rock goby	Seabed, on hard surfaces	Its pelvic fins are fused into a sucker for gripping rock. The clue is in the name — a sucker is no use on sand.
① Common octopus	Seabed, in a den	It does need a den, and the farm's structure and dropped shell supply plenty. Boosted, not built: octopus are mobile and will happily den in debris or a burrow on open ground.
● Conger eel	In the anchor blocks	An obligate hole-dweller — it needs a hole big enough to reverse into and hold. Worth knowing: that hole is the <i>mooring gear</i> , not the crop, so a conger's den outlives the harvest.
○ Lesser-spotted catshark	On the seabed	A sandy-ground shark that curls up on open bottom quite contentedly. No structure required.
○ Grey seal	At the surface	Ranges tens of kilometres between haul-outs. The farm is a snack stop.

THE SAME MAT AFTER THE FARM GOES

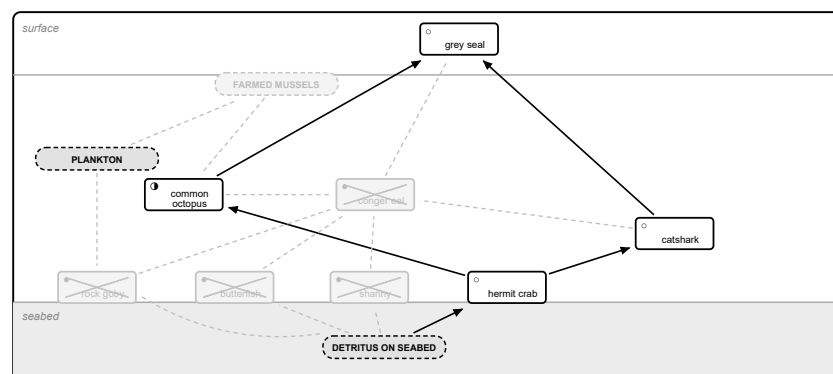
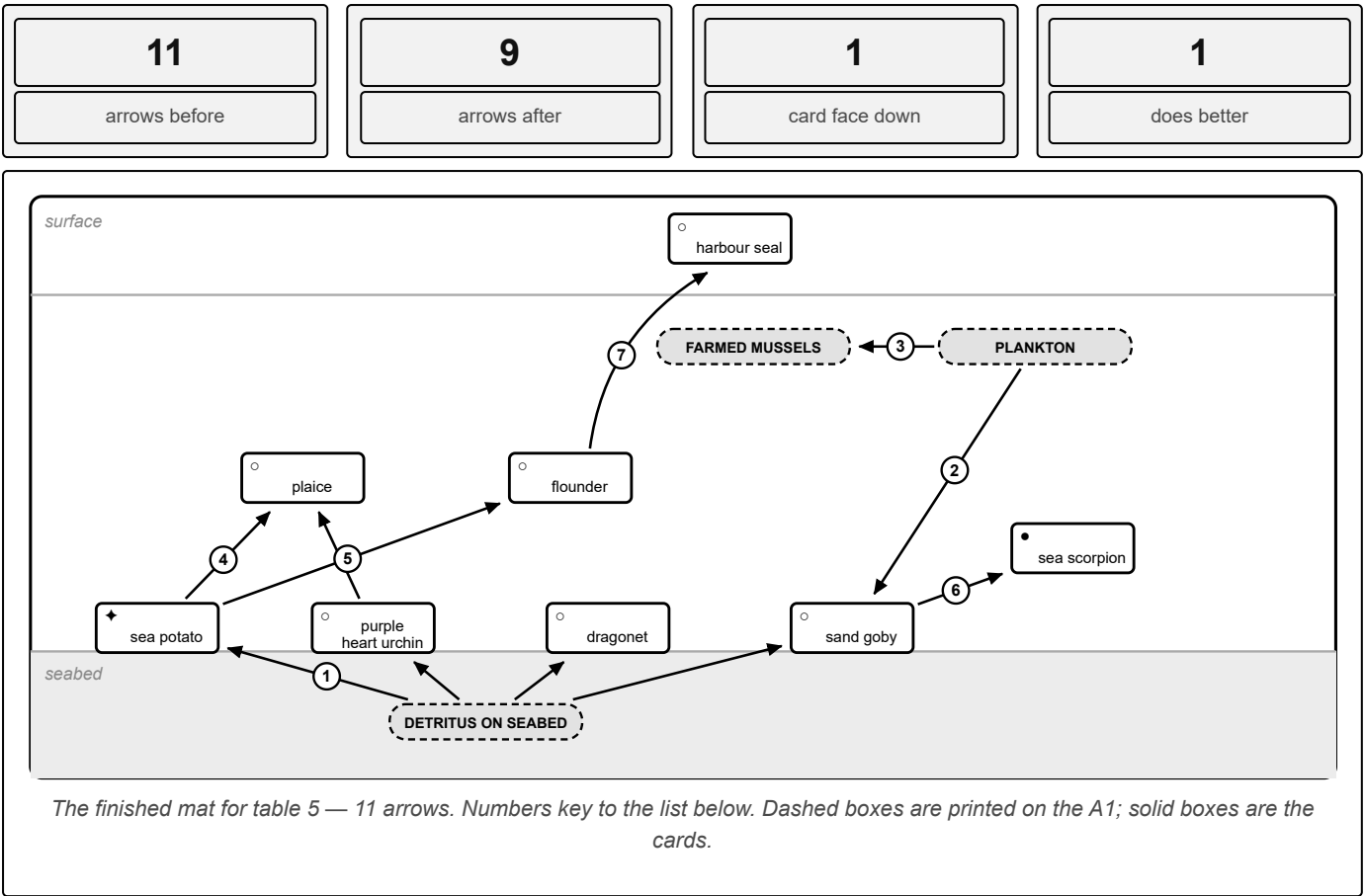


Table 4 after the reveal — 17 arrows down to 5. Crossed boxes are face down, faint dashed lines are the 12 arrows the table crosses out.

● THE TEACHING POINT

The seabed already had detritus before anyone put a farm above it. What it did not have was anywhere to hide, and three of the four cards that go face down are small fish that need a crevice. This is the table that makes the abstract point concrete: *habitat is not food*. If someone says "surely the seabed was fine before", agree — it was fed. It just had no cover.

The control table. This is roughly what the seabed looks like anyway, and it is the single most important deck in the room for your credibility.



WHERE EACH CARD GOES, AND WHY

CARD	PLACE IT	THE REASON
♦ Sea potato	Buried in clean sand	A heart urchin living 10 to 20cm down, sieving organic matter from clean, well-sorted sand. Heavy fallout from a farm clogs and enriches that sediment, which suits it <i>worse</i> . The one card that does better with no farm.
○ Purple heart urchin	Just under the sand	Same feeding guild as the sea potato but tolerant of coarser, gravellier ground, so the enrichment bothers it far less. Roughly unchanged either way.
○ Dragonet	On the sand	Picks worms, small crustaceans and brittlestars off the surface. A sand specialist by trade.
○ Sand goby	On the sand	Near-transparent and vanishes against sediment. Its camouflage only works on sand, so sand is where it stays.
● Long-spined sea scorpion	On hard cover	The exception on this table: an ambush predator whose camouflage matches <i>rock</i> , not sand. It needs hard structure to sit against, so the farm is what puts it here.
○ Plaice	On the sand	A flatfish that eats buried urchins and brittlestars. It belongs on clean sediment.
○ Flounder	On the sand	Another flatfish, and unusually tolerant of fresh water, so it works estuaries too.
○ Harbour seal	At the surface	Mobile, and more of an estuary and sandbank animal than the grey seal.

THE SAME MAT AFTER THE FARM GOES

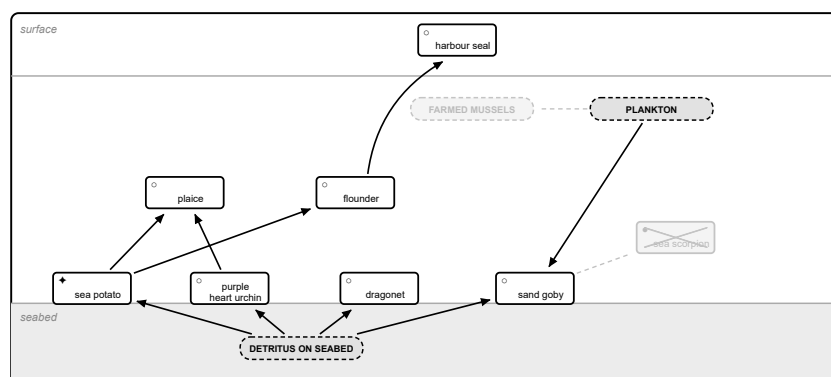


Table 5 after the reveal — 11 arrows down to 9. Crossed boxes are face down, faint dashed lines are the 2 arrows the table crosses out.

● THE TEACHING POINT, AND WHY THIS TABLE IS THE MOST VALUABLE ONE YOU HAVE

Eleven arrows down to nine. Almost nothing happens, and one animal is actively better off. Do not soften that — lead with it. A session made only of tables 1 and 4 would be a sales pitch; table 5 is what makes the other four believable. When the sea potato is held up, the sentence to reach for is: *"a farm is not free. It changes the seabed underneath it, and that suits some things less. We would rather tell you that than have you find out."*

“ SAY ROUGHLY THIS

"Keep your card. The code on the back opens the footage our cameras are collecting on real farms. If you want to know what actually lives on one, you can sit at home and help us identify it. That is not a gimmick, we genuinely use the identifications."

Leave five minutes of slack at the end. You will use it. The best conversations of the hour happen after you have officially finished, usually with the person who said the least.

IF IT GOES WRONG

PROBLEM	WHAT TO DO
Only 12 people come	Run tables 1, 4 and 5 only. That keeps the wipeout table, the seabed table and the sceptic table. The reveal still works.
50 people come	Deal the spare deck as a sixth table and pair people on cards. Do not add a facilitator job you do not have.
No power or wifi	Clips are already downloaded on the laptop. If the laptop dies, use the printed stills and describe the footage. The activity does not depend on the screen.
Running 10 minutes late	Cut step 5 to three minutes and take questions in the slack. Never cut step 4.
A table finishes very early	"Find the longest food chain on your mat." Then, "which of you would still be here without the farm? Predict it before we do the reveal."
Someone dominates	"Let me get a couple of other views and come back to you." Then actually come back to them.
Nobody talks at all	Go table by table rather than asking the open room. Ask the table, not an individual.

AFTERWARDS, FIVE MINUTES WHILE YOU PACK UP

- ☐ Photograph one completed mat before wiping it
- ☐ Note the two questions you could not answer well
- ☐ Rough headcount and how many cards went home
- ☐ Wipe mats before the pens set

• THE MEASURE OF SUCCESS

Not how many people came. Whether the most sceptical person in the room left still talking to you. If a local fisherman filled in the costs column and stayed behind to argue about siting, that was a good workshop.

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 Species, diets and farm impact generated from the FishSpotter catalogue and the PEBL farm food web. A teaching schematic, not a quantified survey.